Linguistics 478/578
Spring '16

# Acoustics

## A Matlab basics

1. Sequences/vectors/arrays
   You can designate a sequence of numbers in several different ways. For example: `[1 4 7]`, `5:12`, `5:.3:11`.

2. Variables
   You can store things under names you designate. For example: `a = [7 6 2]`, `myhat = 1:1000`, `onion = 3:3:27`.

3. Semicolon
   Some commands have *values*: `7+2`. Some commands have *side effects*: `disp('wow')`. Some have both: `x = 7+2`. The value of a command is printed unless you end the command with a semicolon: `x = 7+2` vs. `x = 7+2;`.

4. Indices
   You can refer to the elements of a sequence with indices, which go after the sequence name in parentheses: `x(2)`. Note that indexing starts at 1. Note also that a set of indices can be indicated with a colon: `a(2:3)`.

5. Help
   You can get help on a command `blorg` by typing `help blorg` or `doc blorg` at the command-line or by using the *Search Documentation* box in the upper right of the matlab window.

6. m-files
   You can store a sequence of commands in an m-file. You create and edit those files with the `edit somefilename` command. The contents of an m-file in the current directory can be executed just by typing the name of the file. (You can also write functions and whole programs with m-files if you are so inclined.)

## B Basic waves

7. Sine waves

```
x = 0:.1:30;
s = sin(x);
plot(s)
```

8. Sine waves can vary in frequency:

```
plot(sin(3 * [0:.1:26]))
hold on
plot(sin(2 * [0:.1:26]),'r')
plot(sin(0:.1:26),'g')
hold off
```

9. ... and amplitude:

```
plot(3 * sin([0:.1:26]))
hold on
plot(2 * sin([0:.1:26]),'r')
plot(sin(0:.1:26),'g')
hold off
```

10. To get one cycle of a sine wave, we must go from 0 to $2\pi$. To get to two cycles, we must go from 0 to $4\pi$ in the same distance.

```
plot(sin([0:.1:(2*pi)]))
hold on
plot(sin(2*[0:.1:(2*pi)]),'r')
```

11. Listening to sine waves:

```
x = 5 * sin([0:.1:(100*pi)]);
sound(x,length(x))
```

12. Combining sine waves:

```
x = 2 * sin([0:.1:(4*pi)]);
y = sin(2 *[0:.1:(4*pi)]);
plot(x)
hold on
plot(y,'r')
plot(x + y, 'g')
hold off
```

13. Playing complex waves:

```
x = 2 * sin([0:.1:(2000*pi)]);
y = sin(2 *[0:.1:(2000*pi)]);
sound(x,length(x))
sound(y,length(y))
sound(x+y,length(y))
```

14. An *extremely* rough approximation of a glottal pulse:

```
z = pulstran(0:1/1e4:1,0:1/100:1,'tripuls',.001,-1);
plot(z);
plot(z(1:1000));
sound(z,10000);
```

15. Using the `spec()` function (on website) on the glottal pulse approximation:

```
plot(spec(z,10000))
```

16. You can look at the filtering effect on vowels with your own speech with the `snd2spec` function (on the website).

17. Average formant values for some American vowels:

| Vowel | Example | F1 | F2 | F3 |
|---|---|---|---|---|
| ɑ | cot | 785 | 1154 | 2598 |
| æ | cat | 754 | 1874 | 2609 |
| ʌ | cut | 689 | 1291 | 2556 |
| ɔ | caught | 576 | 872 | 2555 |
| ɛ | bet | 563 | 2074 | 2718 |
| r̩ | curt | 495 | 1488 | 1831 |
| ɪ | bit | 414 | 2213 | 2795 |
| i | beet | 286 | 2518 | 3109 |
| ʊ | good | 451 | 1086 | 2447 |
| u | boot | 339 | 914 | 2435 |

18. Generating and filtering noise. The simplest way to do this is to use the `fdatool` command. (There's a starter eaxmple on website: `mhfdatoolexample.fda`) You can also create filters by hand with the `fdesign` command. Here are some examples.

```
x = wgn(10000,1,3);
hf1 = design(...
   fdesign.bandpass(...
      'N,F3dB1,F3dB2',10,1000,2000,10000));
hf2 = design(fdesign.highpass('N,Fc',10,2500,10000));
y1 = filter(hf1,x);
y2 = filter(hf2,x);
subplot(3,1,1);
bar(spec(x,10000));
subplot(3,1,2);
bar(spec(y1,10000),'r');
subplot(3,1,3);
bar(spec(y2,10000),'g');
sound(x,10000)
sound(y1,10000)
sound(y2,10000)
```